
Lab prompts — Sessions 8 & 9

Géron Chapters 1 and 2

Computing, Robotics & Bionics • 3rd-year Bioengineering • 5 & 8 October 2026

These prompts run on *Géron's own notebooks*, from the PyTorch edition's repository `ageron/handson-mlp`. Running a notebook end to end teaches less than it feels like it does, because nothing in it resists you. Each prompt below asks for a change you have to reason about first, and each one names **what to hand in** — usually one number, one plot, or one sentence. None of them can be answered by re-running a cell.

Key idea

Ground rules. Open the notebook in Colab and immediately *Save a copy in Drive*; work on your copy, not on Géron's. Keep each answer in a new cell directly below the one you modified, with a markdown heading naming the prompt. Hand in the single notebook per chapter, with your name in the first cell. If you are stuck on the *why* rather than the *how*, the section of the Part I companion is named at the end of each prompt.

1 Chapter 1 — 01_the_machine_learning_landscape.ipynb

Four prompts — about 30 minutes

1. **Swap the model, keep the data.** The notebook fits a `LinearRegression` to life satisfaction against GDP per capita. Refit the same data with `KNeighborsRegressor(n_neighbors=3)` and predict for Cyprus (GDP per capita $\approx$ \$37,655). *Hand in:* the two predictions, and **one sentence** saying which you would trust and why — argued in terms of instance-based versus model-based generalisation, *not* in terms of which number looks nicer. (Companion §1.1)
2. **Make the training data unrepresentative, on purpose.** Refit the linear model using only the countries in the middle of the GDP range — drop everything below \$23,500 and above \$62,500 — then predict for one very poor and one very rich country that you dropped. *Hand in:* the two predictions from each model and the gap between them, plus **the name of the challenge** from Chapter 1 that you have just demonstrated. (Companion §1.2)
3. **Add a feature that cannot possibly help.** Add a second input column filled with random numbers (`np.random.rand`) and refit the linear model. *Hand in:* the training error before and after, the Cyprus prediction before and after, and **one sentence** explaining why training error alone is never evidence that a model improved. (Companion §1.2–1.3)
4. **Overfit deliberately.** Fit a degree-10 polynomial to the same data. Then split the countries into a training half and a held-out half and refit both the linear and the degree-10 model on the training half only. *Hand in:* a table of four numbers — training and held-out error for each model — and **one sentence** on which of the four you would quote to a colleague. (Companion §1.2)

2 Chapter 2 — 02_end_to_end_machine_learning_project.ipynb

Before you start

This notebook is long. Do not try to finish it in one sitting: run it as far as *Explore and Visualize the Data* first, then come back for the rest. The prompts below are placed in the order the notebook reaches them.

Four prompts — about 45 minutes

1. **Stratify on something else.** Géron stratifies the split on `income_cat`. Build a different stratification variable — bands of `housing_median_age`, or `ocean_proximity` as it stands — and reproduce his comparison table of category proportions for a stratified split against a purely random one. *Hand in:* the largest percentage-point error in each of the two columns, and **one sentence** on which attribute is worth stratifying on and how you would decide that for a clinical dataset. (Companion §2.2)
2. **Invent an attribute he does not try.** The notebook builds `rooms_per_house`, `bedrooms_ratio` and `people_per_house`. Invent one more combination that he does not try. State your physical reasoning for why it should carry signal *before* you compute anything, then compute its correlation with `median_house_value`. *Hand in:* your reasoning in one sentence, the correlation, and whether it beats `bedrooms_ratio`. An honest negative result is a full answer here. (Companion §2.3)
3. **Commit the leak on purpose, then measure it.** Standardise the *whole* dataset

before splitting, train the linear model, and record the cross-validated RMSE. Now do it properly — split first, fit the scaler on the training folds only — and record it again. *Hand in:* the two numbers and their difference, and **one sentence** on what it means that the difference is small. (This is the point of the exercise: the leak is cheap to commit and almost invisible in the score, which is exactly why the rule has to be procedural rather than a judgement call.) (Companion §2.3)

4. **How much is your single number worth?** Re-run the train/test split with three different values of `random_state`, retraining and re-scoring each time. *Hand in:* the three test RMSEs and their spread, and **one sentence** on what that spread implies about quoting a single test score to three decimal places. (Companion §2.2, §2.4)

If you have time

Re-score the final model with MAE as well as RMSE and report both. Say which is larger, why it must be larger on this dataset, and which one you would put in front of a non-technical stakeholder. (Companion §2.1)

What this is preparing you for

Miniproject 1 is released at the end of Session 9 and follows the Chapter 2 workflow step for step: frame the problem, choose a measure, freeze a test set, prepare the data, select a model, validate once. Every prompt above is one of those steps done deliberately wrong or deliberately differently, so that you can see what the step is actually protecting you from. Bring your two notebooks to the Session 10 catch-up if any answer surprised you.